

International Conference on Advancement in Cyber Security and Digital Forensics



Special Session: Cyber Resilient Autonomous EVs

Overview

The electrification of transportation is undergoing a paradigm shift with the integration of Artificial Intelligence (AI) and Machine Learning (ML). Modern electric vehicles are no longer passive energy systems but intelligent cyber-physical entities capable of perception, prediction, adaptation, and autonomous decision-making. The session bridges AI, control systems, cybersecurity, and sustainable mobility.

Topics of Interest

- AI/ML for Battery Management Systems (BMS)
- Deep Learning for SoC, SoH, and RUL Estimation
- Reinforcement Learning for Energy Management in EVs
- AI-based Fault Detection and Predictive Maintenance
- Digital Twin Frameworks for Electric Vehicles
- Autonomous Driving Algorithms in EV Platforms
- Cybersecurity in EV Networks and Charging Infrastructure
- AI-enabled smart charging and V2G Systems
- Edge AI for Embedded EV Systems
- Federated Learning for Distributed EV Intelligence
- Explainable AI (XAI) in Safety-Critical EV Applications
- AI-driven Thermal Management in Batteries
- Intelligent Power Electronics and Converter Control

Session Organizers



Dr. Souvik Ganguli

Thapar Institute of Engineering and Technology, Patiala, Punjab

Email: souvik.ganguli@thapar.edu



Dr. Srabanti Maji

DIT University, Dehradun

Email: srabanti.maji@dituniversity.edu.in